SAT Hard Question Set 2

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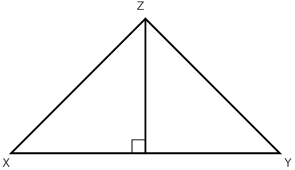
Phase 2 · Hard Question Practice

Overview

Question 1



In the figure shown, the measure of angle Y is 38°. The length of XY is 12 meters, and the length of YZ is 5 meters. What is the area, in square meters, of triangle XYZ?

A. 30

B. 60

C. $30 \sin 38^\circ$

D. $60 \sin 38^\circ$

Answer 1

Correct Answer: C

Question 2

A research manager selected two random samples to estimate the average lifespan of a certain type of light bulb (in hours). Based on the first sample, the research manager estimated that the average lifespan of this type of light bulb is 800 hours, with a margin of error of 20 hours.
Based on the second sample, the research manager estimated that the average lifespan of this type of light bulb is 810 hours, with a margin of error of 35 hours. Assuming the margins of error were calculated in the same way, which of the following best explains why the first sample obtained a smaller margin of error than the second sample?

A. The first sample contained fewer light bulbs than the second sample.

B. The first sample contained more light bulbs than the second sample.

C. The first sample had a shorter average lifespan than the second sample.

D. The first sample had a longer average lifespan than the second sample.

Answer 2

Correct Answer: B

Question 3

The function $p(x)$ is defined by:

$$p(x) = a((x-3)^2 + b)((x-3)^2 - c),$$

where a, b, and c are constants. The graph of $y = p(x)$ passes through the points $(1, 25)$ and $(-1, 52)$.
What is the value of $p(5) + p(7)$?

A. 25

B. 27

C. 52

D. 77

Answer 3

Correct Answer: D

Question 4

In the given function, b is a positive constant. For every increase in the value of x by 1, the value of $f(x)$ increases by $c\%$, where $0 < c < 100$.
Which expression gives the value of c in terms of b?

$$f(x) = 77(b)^x$$

A. $1 + \frac{b}{100}$

B. $b + 100$

C. $100(b + 1)$

D. $100(b - 1)$

Answer 4

Correct Answer: D

Question 5

On average, a certain tree grows 87 centimeters every m months. At this rate, which expression represents the number of centimeters, on average, the tree grows every k years?

A. $\dfrac{29m}{4k}$

B. $\dfrac{29k}{4m}$

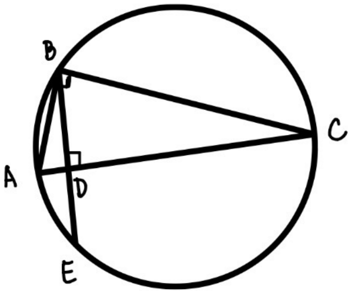
C. $\dfrac{1044m}{k}$

D. $\dfrac{1044k}{m}$

Answer 5

Correct Answer: D

Question 6



In the figure shown, points A, B, C, and E lie on the circle, and $AB < BC$. Segment AC is perpendicular to segment BE at point D, and $BD = \sqrt{346}$. The diameter of the circle is 175. If $\frac{CD}{AD} = r$, what is the value of r?

Answer 6

Final Answer: $\boxed{86.5}$

Question 7

In the given equation, a and k are constants, where $k > 5a$. The sum of the solutions to the equation is $3k + 32$. What is the value of a?

$$(x - k)^2 = (k - 5a)(x - k)$$

A. $-\dfrac{32}{5}$

B. $\dfrac{32}{5}$

C. $-\dfrac{5}{32}$

D. $\dfrac{5}{32}$

Answer 7

Correct Answer: A

Question 8

The equation $N(m) = 63(Q)^{m/6}$ gives the predicted population $N(m)$, in thousands, of a certain bacteria colony m minutes after the initial measurement, where Q is a constant greater than 1. The predicted population increases by $p\%$ every 180 seconds.
What is the value of p in terms of Q?

A. $100(Q^{30} - 1)$

B. $100(Q^{\frac{1}{2}} - 1)$

C. $100(Q^{30} + 1)$

D. $100(Q^{1.5} + 1)$

Answer 8

Correct Answer: B

Question 9

In the given equation, m and n are constants. The sum and product of the solutions of the equation are 8 and $\frac{7}{2}$, respectively.
What is the value of $\frac{m}{n}$?

$$(x - m)^2 = (m + 2n)(2x - n)$$

A. $-\frac{1}{3}$
B. $\frac{1}{3}$
C. -3
D. 3

Answer 9

Correct Answer: D

Thank You!

We appreciate your time and focus.

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